

Pakhi Banchalia

Dresden, Germany | pakhibanchalia24@gmail.com | [GitHub](#) | [Portfolio](#)

Education

- **Technische Universität Dresden** Oct 2025 - Present
Master of Science in Computational Modeling and Simulation (Applied AI track) Dresden, Germany
- **Amrita Vishwa Vidyapeetham** 2021-2025
Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) Amritapuri, India

Experience

- **Aalto University** Jun 2026 - Aug 2026
Summer Research Intern Espoo, Finland
 - One of 67 interns selected from 5,500 applications across 133 countries for the AScI International Summer Research Programme.
 - Picked up empowerment-based assistance in multi-agent reinforcement learning within weeks and proposed experimental directions in regular discussions with supervisors.
 - Found that AvE (NeurIPS 2020) tests assistance with an agent acting remotely rather than embodied; rebuilt the experiment with an embodied assistant to measure what that costs.
- **Lossfunk** Jul 2025 - Sep 2025
Research Intern Bangalore, India
 - Built and evaluated homeostatic reinforcement-learning agents in the Crafter environment, using multiple internal variables such as hunger, energy, and safety to shape agent behaviour.
 - Tested how alternative reward formulations and competing drives affect exploration, survival, behavioural diversity, and policy stability, against standard RL baselines.
- **Computational Machinery of Cognition (CMC) Lab, TU Dresden** Nov 2024 - Present [Intermittent]
Research Intern (Independent Collaboration) Dresden, Germany
 - Designed RL agents with internal physiological drives to study retention and generalization across changing task distributions.
 - Analysed neural and behavioural data from trained foraging RL agents, decoding movement and internal state from network activity and linking oscillating unit groups to gait.
 - For a separate project, implemented a reward-modulated, gradient-free learning algorithm based on weight perturbations to investigate biologically plausible credit assignment without backpropagation.
- **Google Summer of Code, INCF** May 2024 - Sep 2024
Contributor Remote
 - Simulated *C. elegans* neural development using Neural Cellular Automata and evolutionary optimization, representing developmental changes in cell state and connectivity over time.
 - Modeled evolving neural connectivity with Graph Neural Networks and performed Topological Data Analysis with Kepler Mapper to inspect developmental and connectivity trajectories.
 - Delivered reproducible, documented open-source research code for computational developmental neuroscience. [\[Code\]](#)

Leadership

- **amFOSS** Sep 2022 - Jul 2025
Core Committee Member; Head of Design Amritapuri, India
 - On a five-person core committee for 50-55 active members: ran weekly project check-ins, surfaced stalled work, and bridged the founder, university staff, and members.
 - Led the design team, reviewing work and making final design decisions, including the amfoss.in redesign.
 - Mentored junior cohorts and ran performance reviews tied to continued membership.

Publications

- Pillai, H.S., **Banchalia, P.**, Jamboji, V., Krishnan, S.R. (2026). *CAN: Continuously Adapting Network*. In *Data Science and Applications, ICDSA 2025. Lecture Notes in Networks and Systems*, vol 1723. Springer, Cham. [\[DOI\]](#)

Projects

- **Polaris: Rare-Disease Atlas, Hack-Nation 7th Global AI Hackathon**
 - Mapped rare-disease communities by mechanism and mutation direction for the Buffalo Initiative and OpenAI challenge, starting with seven rare epilepsies; every link cites its source. [\[Live\]](#)
 - Co-developed the idea with a teammate, then built the interface and the researcher view, which screens 2,868 approved medicines by mutation direction.
- **Alignd: Sponsorship Agent for TinyFish Accelerator**
 - One of 60 demos selected from over 2,000 applications to the TinyFish Accelerator, a \$2 million program run by TinyFish and Mango Capital. Reached Phase 2.
 - Defined and built an AI agent that finds and vets YouTube creators for a brand's campaign in six stages, deciding what the agent may conclude alone and what the brand can overrule. [\[Code\]](#)
- **MedX: X-ray Editing Tool for GE HealthCare Precision Care Challenge**
 - Reached the final round of GE HealthCare's Precision Care Challenge 2024 with a team-built web tool that edits X-ray images to generate training data.
 - Owned the Next.js and TypeScript front end, including its visual design, layout and the canvas where a user places an object into an X-ray.
- **SEANN: Self-Expanding and Adaptive Neural Networks (BTech Thesis)**
 - Extended the CAN continual-learning framework, which mitigates catastrophic forgetting by preserving significant task-specific connections through selective neuron freezing and Hebbian learning principles inspired by synaptic plasticity.
 - Introduced an alternative to conventional model scaling that dynamically extends network capacity and identifies task-relevant subnetworks, reducing interference between tasks and catastrophic forgetting.

Talks and Posters

- **Brain-like features in homeostatic deep reinforcement learning agents @ Bernstein Conference 2025**
 - Poster presentation on homeostatic RL agents integrating internal and sensory inputs, showing structured behavioral and neural patterns.
- **Emergence of Brain-like Features in Homeostatic RL Agents @ BrainBody 2025**
 - Oral presentation; demonstrated coupling between internal states, neural dynamics, and adaptive exploration.

Skills

- **Programming Languages:** Python, C++, C, Java, JavaScript, TypeScript, SQL, MATLAB, HTML, CSS
- **Machine Learning and AI:** PyTorch, TorchVision, Gymnasium, JAX, Reinforcement Learning, Multi-Agent Reinforcement Learning, Continual Learning, Graph Neural Networks, Neural Cellular Automata, AI Agents, LLM APIs, Anthropic API, Gemini API
- **Data Science:** NumPy, SciPy, Pandas, Matplotlib, Kepler Mapper, Weights and Biases
- **Software Development and Design:** FastAPI, Django, Next.js, React, n8n, Git, Linux, Docker, Figma